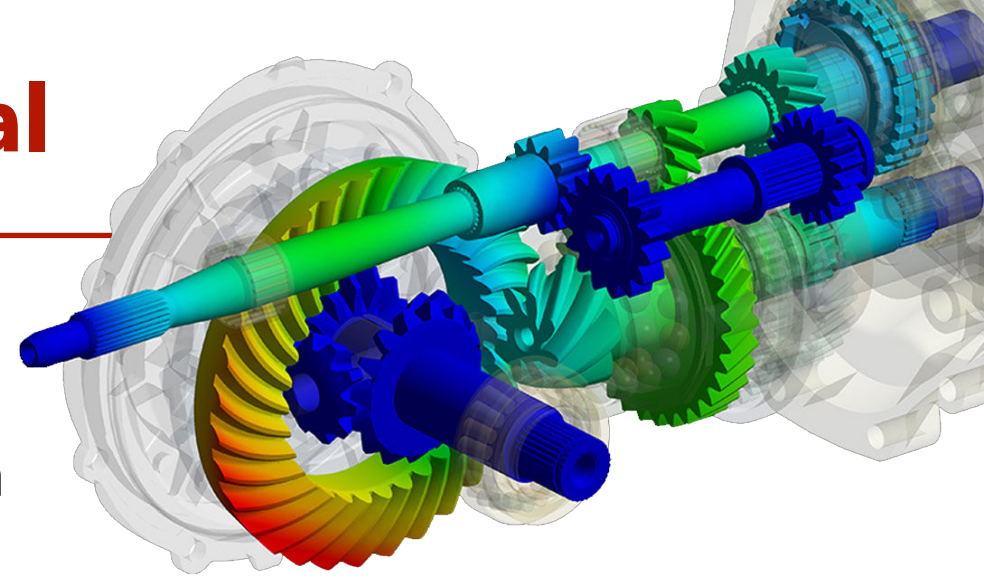


Tutorial on Ansys Mechanical

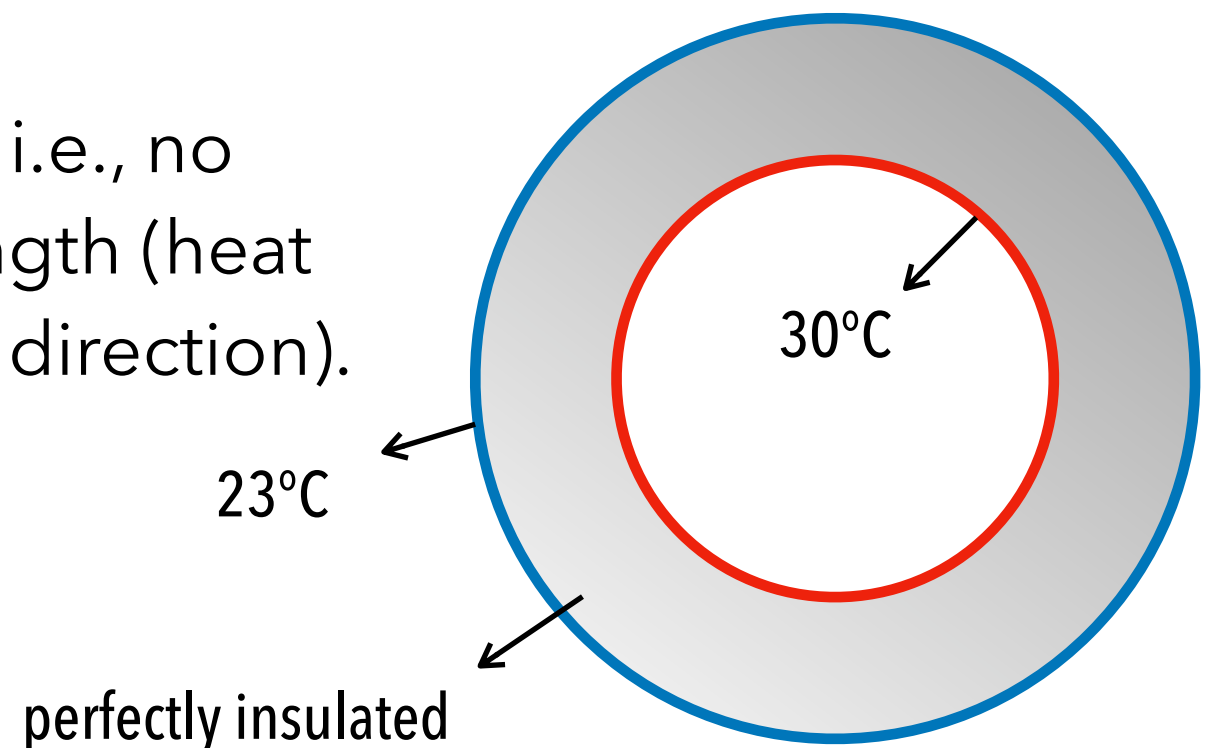


- ▶ ANSYS is engineering simulation software suite used to predict how products function in the real world by creating a **virtual prototype**.
- ▶ ANSYS Mechanical is a specific tool within the suite used for structural, thermal, and acoustic analysis.
- ▶ It uses **finite-element analysis (FEA)** by breaking complex geometries into small, solvable shapes (elements) connected by points (nodes).
- ▶ **Why use it?** To solve partial differential equations (like the Heat Flux Equation) over complex shapes where hand calculations are impossible.

$$\vec{q} = -k \nabla T$$

Steady-state thermal analysis of a cylinder

- ▶ In this recitation, we will focus on heat transfer and thermal analysis of cylindrical objects using a CAD/FEA software.
- ▶ **Problem statement.** We will model the temperature distribution across a solid cylinder made of structural steel.
- ▶ The inner surface of the cylinder will be maintained at an elevated temperature of 30°C , while the outer surface is exposed to ambient conditions at 23°C .
- ▶ We will apply thermal insulation, i.e., no heat flow along the cylinder's length (heat transfer is restricted to the radial direction).

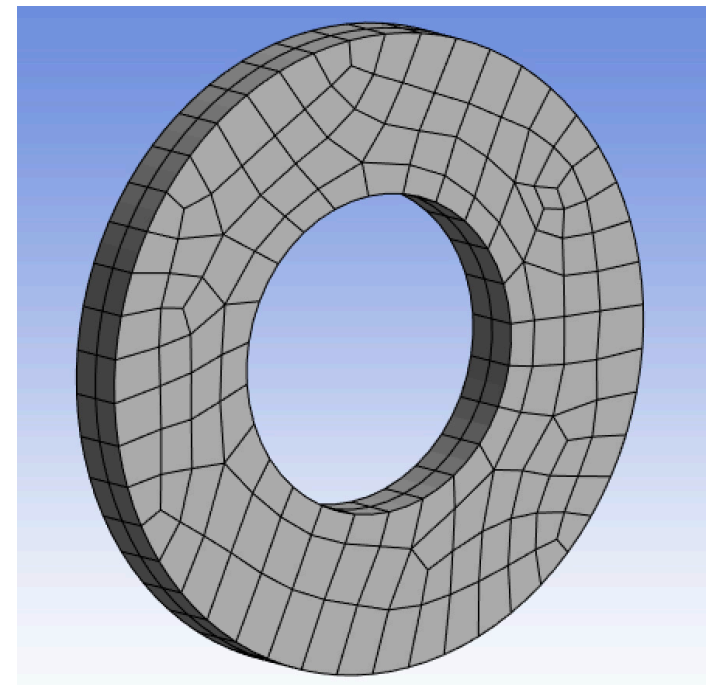
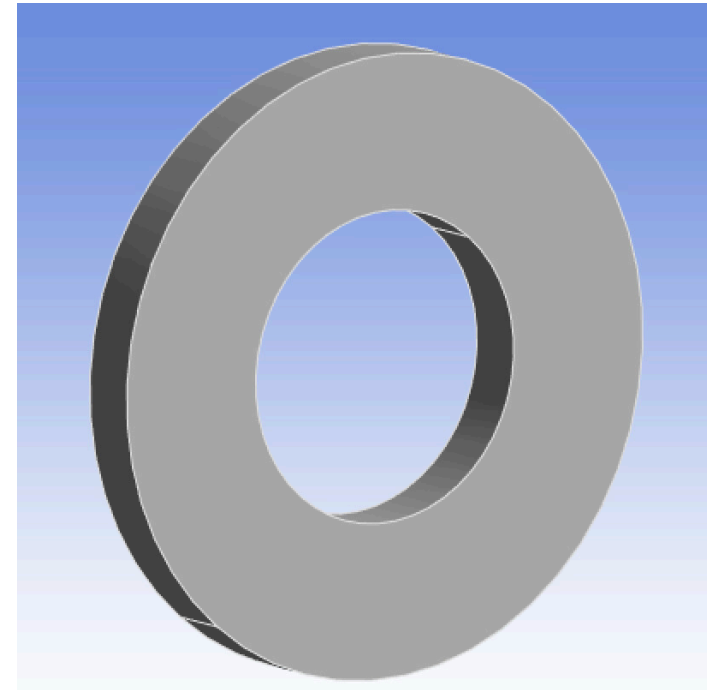


Steady-state thermal analysis of a cylinder

- ▶ You can launch ANSYS via VirtualAndrew.
- ▶ The first step is to start a new project, **Geometry Imports** > **Add a Geometry Import** from files.
- ▶ Next, create a **Steady-State Thermal** analysis, where the temperature distribution does not change with time.

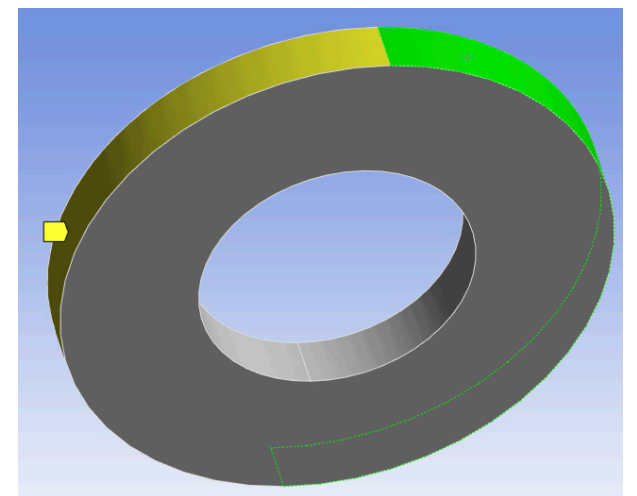
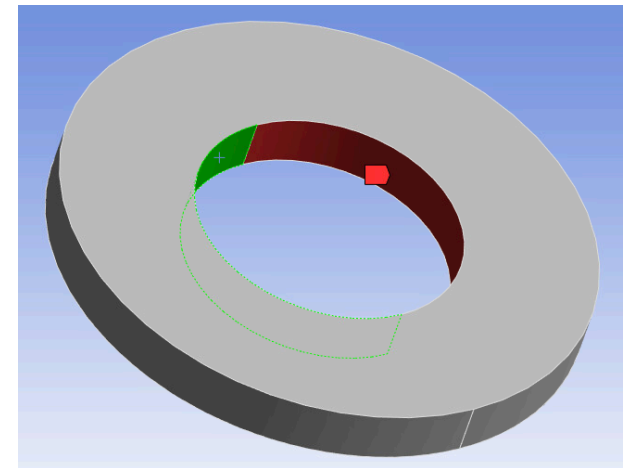
$$\frac{\partial T}{\partial t} = 0$$

- ▶ In **Model** > **Geometry** > **thermal analysis of cylinder**, set the **Material** to Structural Steel.
- ▶ Create a mesh in **Model** > **Mesh** > **Generate Mesh**.



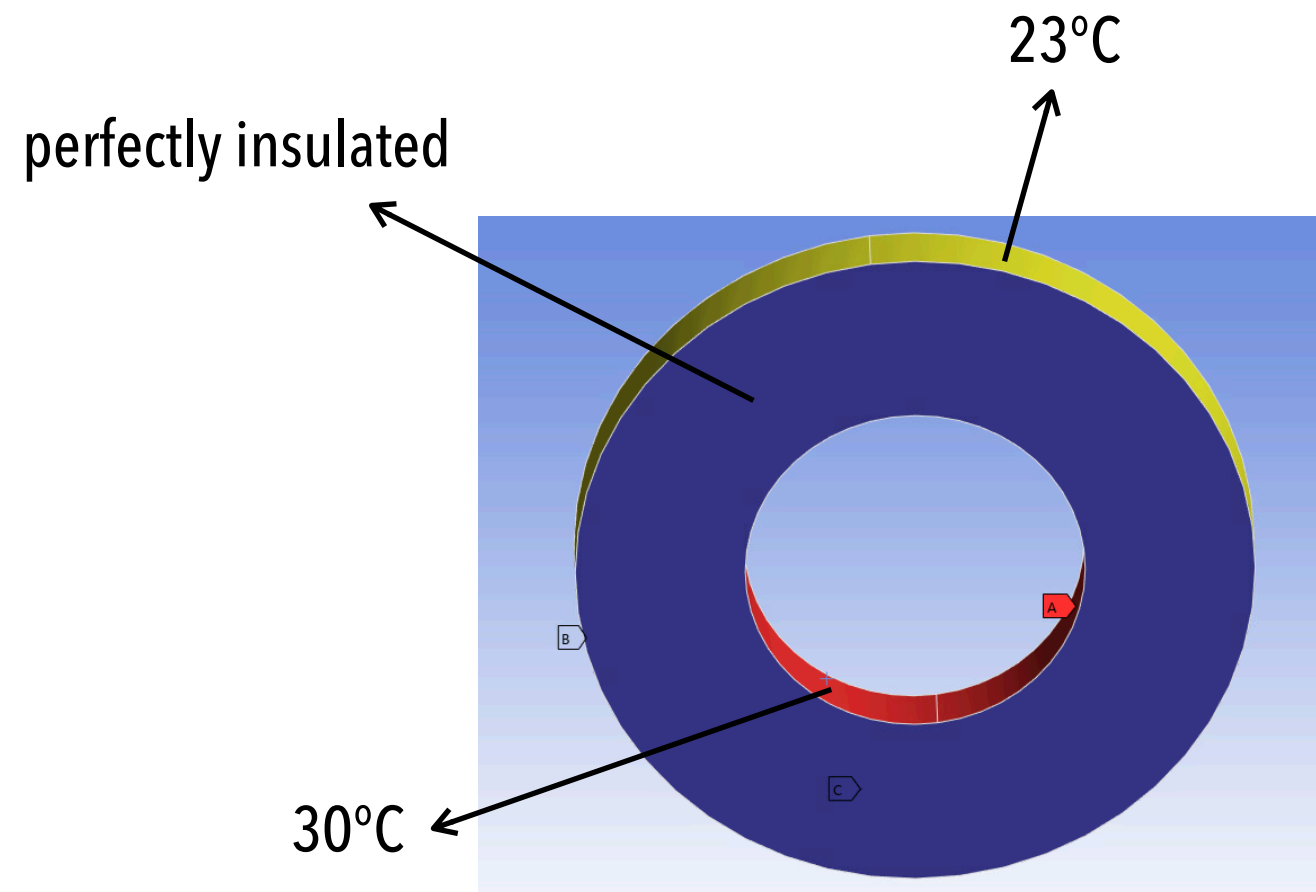
Applying temperature boundary conditions

- ▶ Select **Environment** > **Temperature**, and add a temperature module.
- ▶ You can also modify the measuring units in **Home** > **Units**.
- ▶ To assign temperature, go to **Temperature** > **Scope** > **Geometry**, pick the inner face and apply selection.
- ▶ In **Temperature** > **Scope** > **Definition**, set the **Magnitude** for the interior face to 30°C.
- ▶ To set a convection condition, select the project, select **Environment** > **Convection** > **Scope** > **Geometry**, pick all the outer faces and apply selection.
- ▶ In **Convection** > **Definition**, set the **Film Coefficient** to 50 W/m².°C, and the **Ambient Temperature** to 23°C.



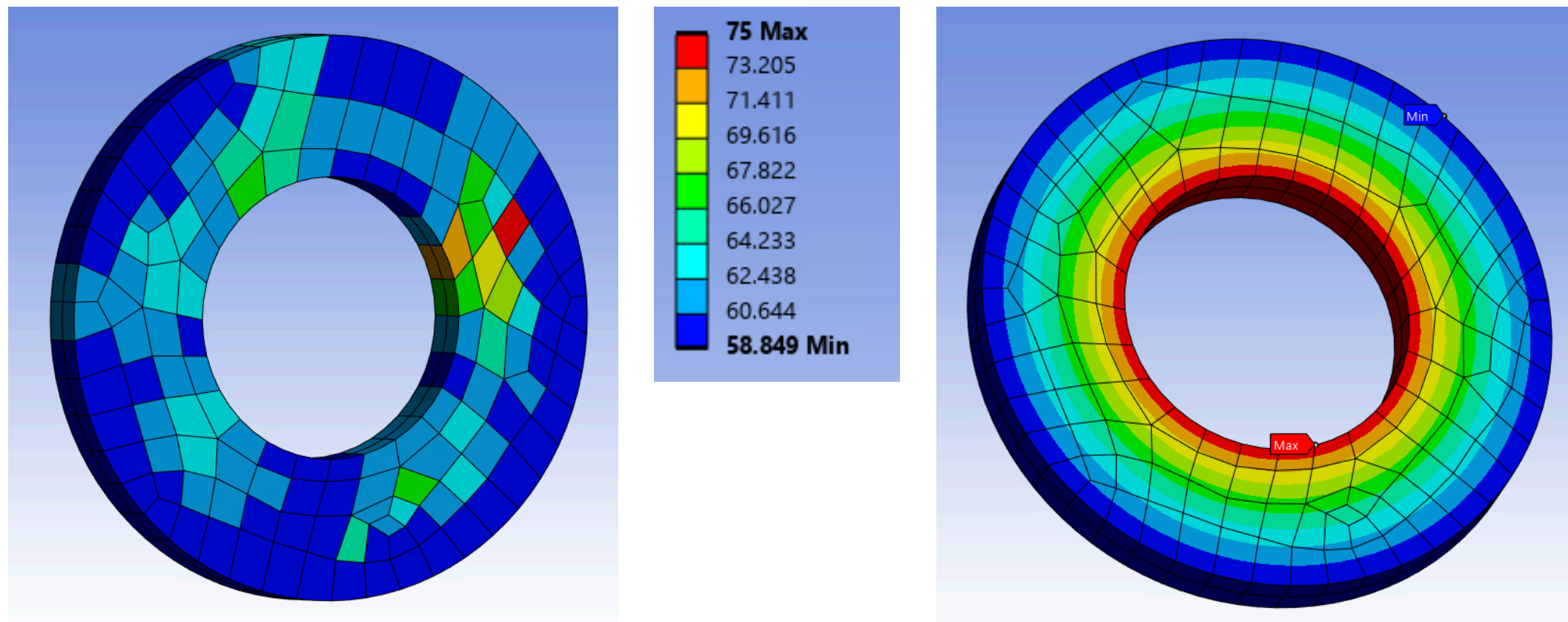
Applying temperature boundary conditions

- ▶ Apply insulated boundary conditions via **Steady-State Thermal > Heat > Perfectly Insulated**.
- ▶ Next, select **Heat Flow > Scope > Geometry**, and select the two flat faces.
- ▶ Select **Steady-State Thermal > Solution > Thermal > Temperature**.



Solving for the temperature distribution

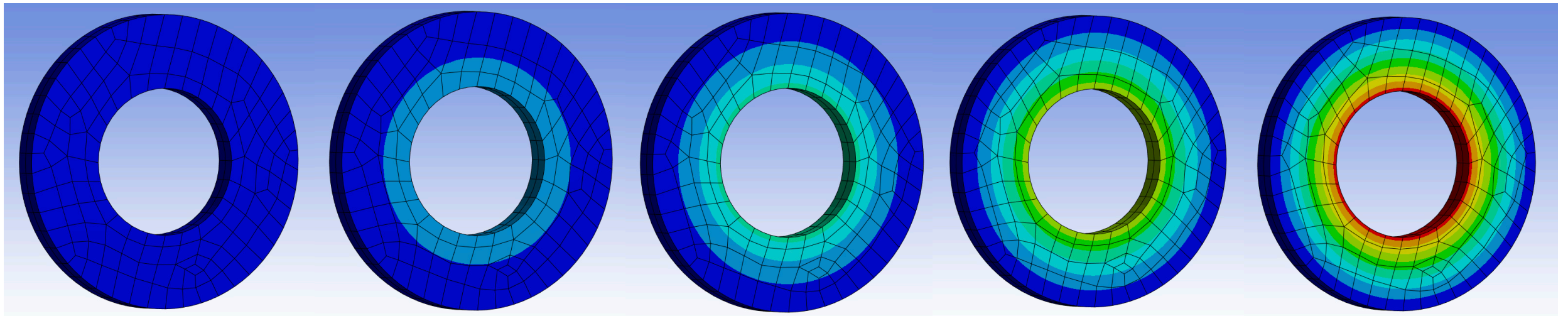
- ▶ Finally, select **Solution** > **Solve**, and wait. :-)
- ▶ This will render the heat flow due to the temperature gradient through the generated mesh in desired geometry.



- ▶ What do you infer from the temperature contours?

Solving for the temperature distribution

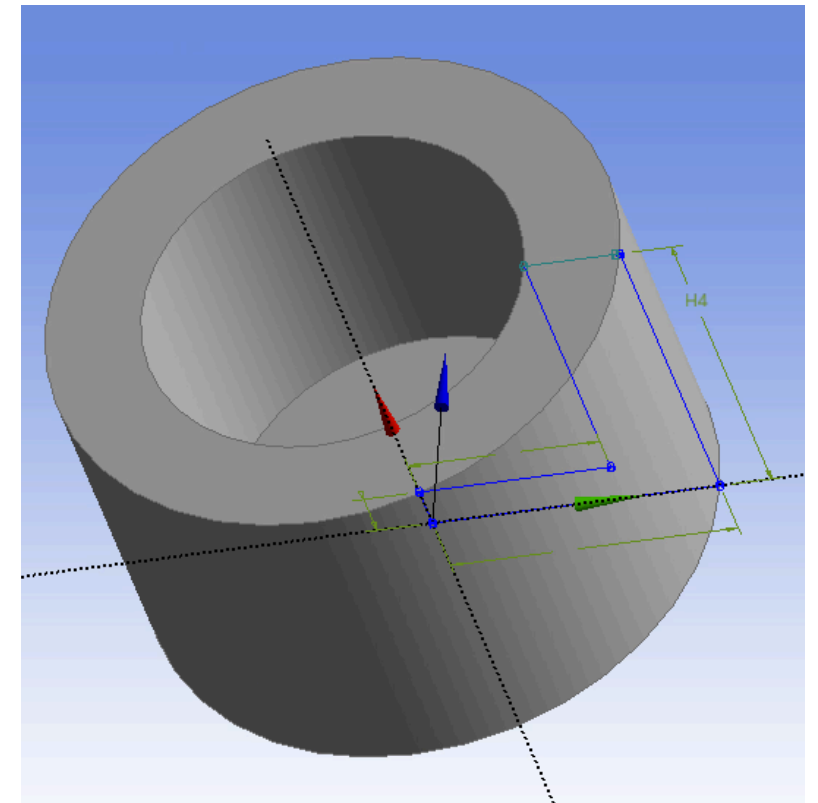
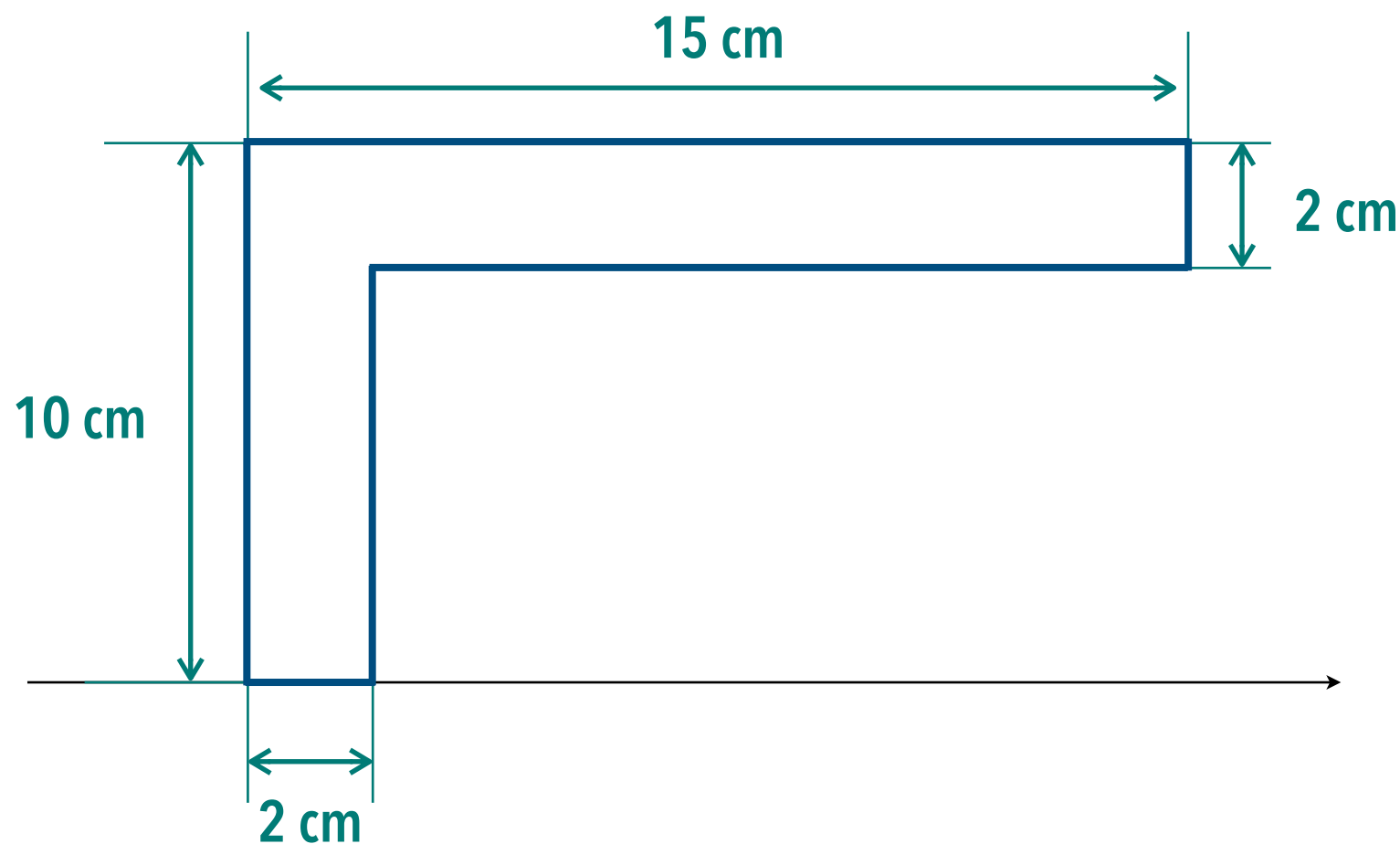
- You can also animate the temperature contours for the heat flow.



- The temperature of the inner face is the same as before, whereas the temperature of the outer face has now increased due to the **conduction heat transfer** between inner and outer edges.
- **Question:** How would increasing the convection heat coefficient value affect the result?

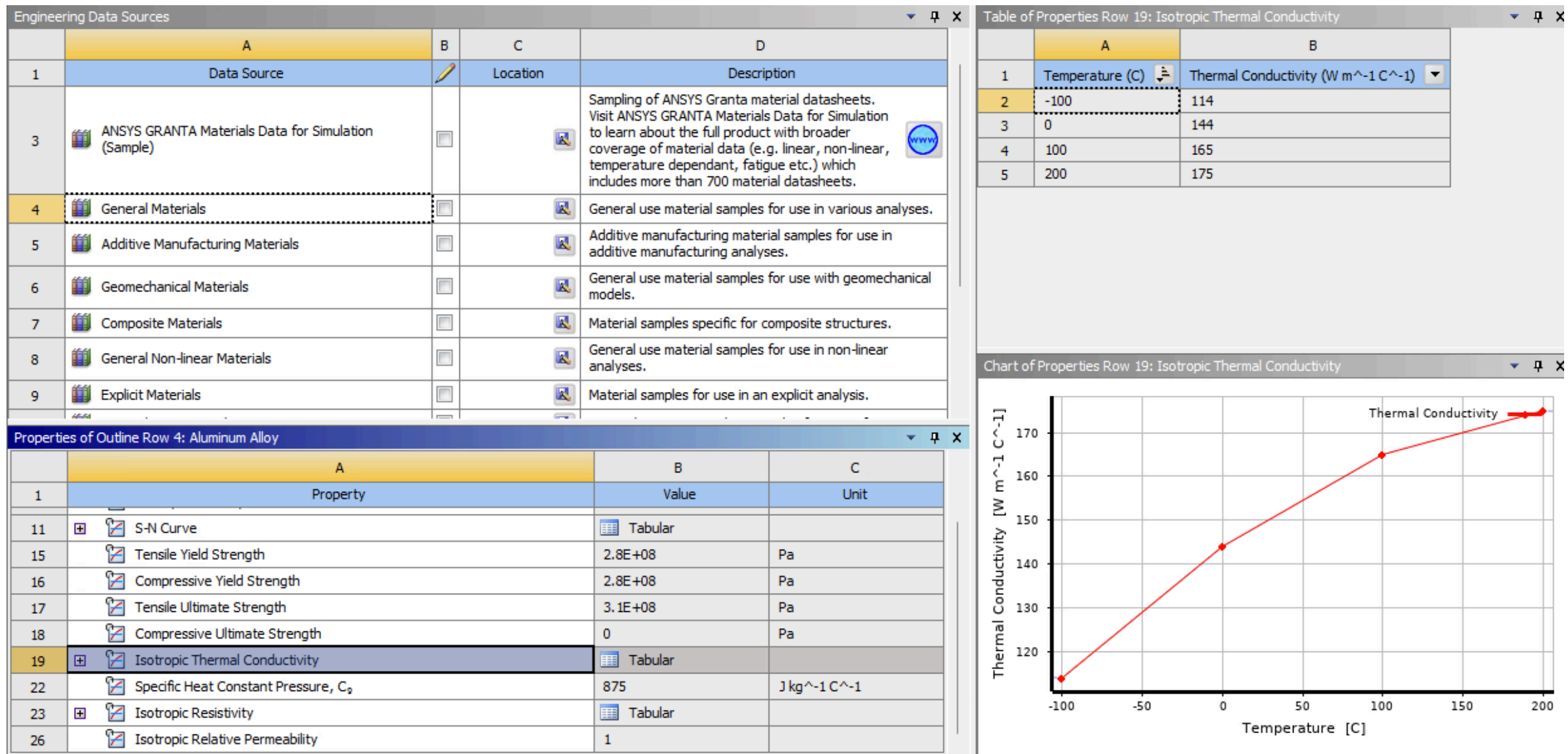
Creating a geometry from scratch

- ▶ We will now utilize ANSYS Workbench, which provides full flexibility over materials selection and generating models.
- ▶ **Problem statement.** We will generate a cylindrical object using the following dimensions for the cross-section, revolving it around the z-axis.



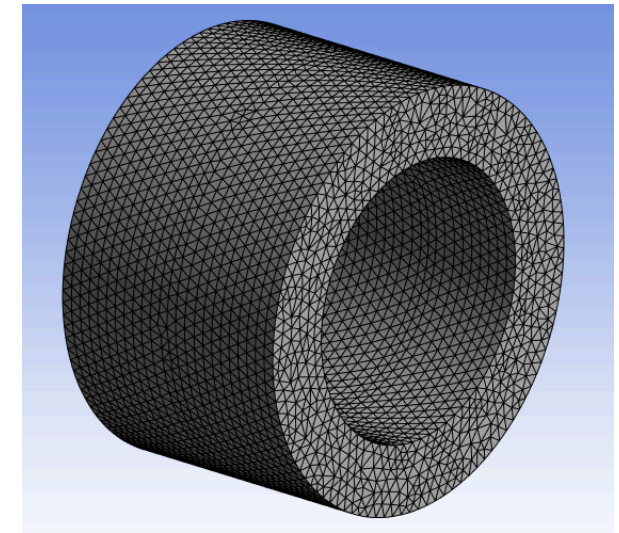
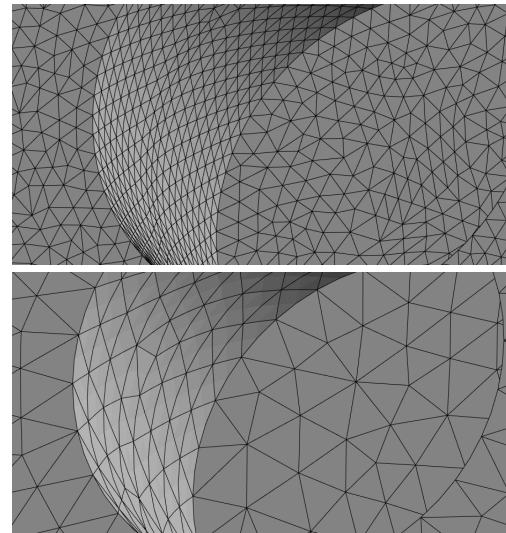
Materials selection

- ▶ You can filter and modify the Engineering Data > Materials.
- ▶ Options are provided for property comparisons among materials.

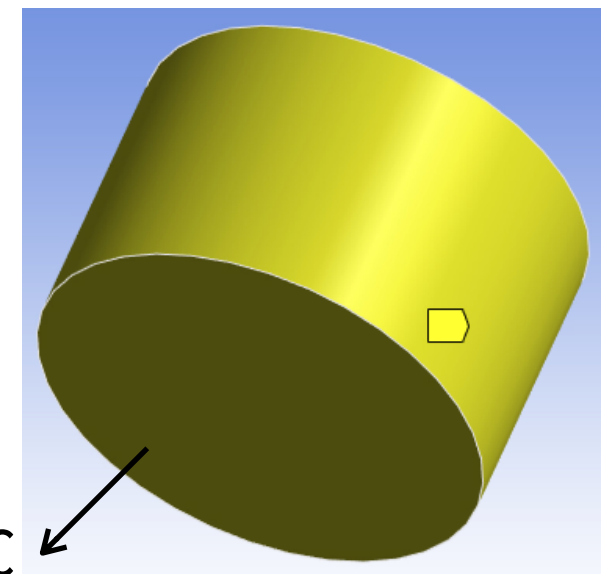
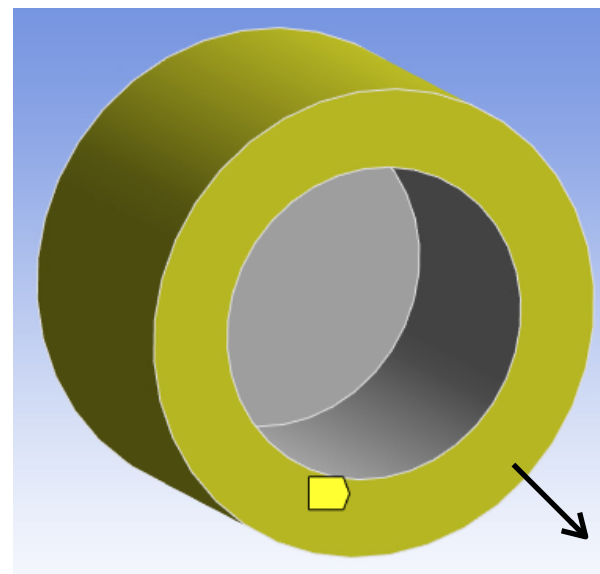
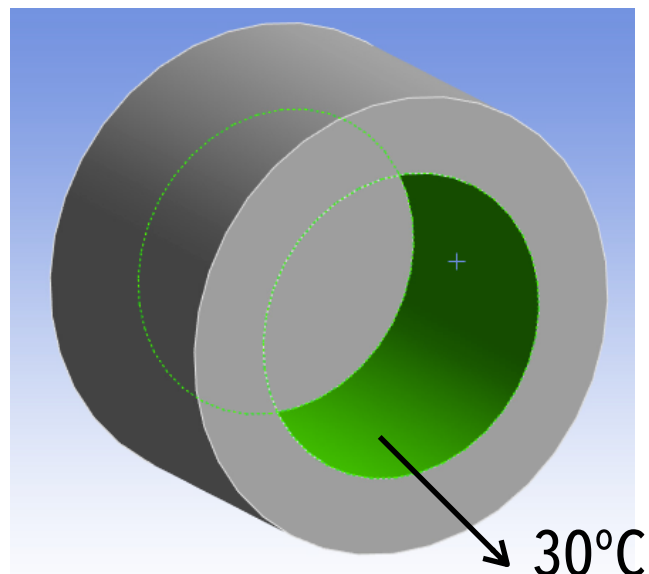


Generating the model

- ▶ Create a mesh and modify the **Sizing** > **Resolution** to make the mesh coarser or finer.



- ▶ The temperature at the inner face is 30°C , and a **heat flux** from the inner edge to the outer edge (coefficient = $200 \text{ W/m}^2\cdot^{\circ}\text{C}$).
- ▶ The outer faces are cooled via **convection** with air with a (coefficient = $50 \text{ W/m}^2\cdot^{\circ}\text{C}$, ambient temperature at 23°C).



Solving for the temperature distribution

- Solve for the heat flux and temperature distribution.
- Result:

